

①

Exact differential equation  
( $x dy + y dx = 0$ )

Sheet - ②

⑫

▣ Prove the necessary and sufficient condition for a differential equation of first degree being exact.

Proof:-

• Necessary condition :- If  $Mdx + Ndy = 0$  is exact,  
then,  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Let Let,  $Mdx + Ndy = 0$  ..... ①

Since, the equation is exact, so it has an exact differential, let that is  $df$ , where  $f$  is a function of  $(x, y)$

$\therefore df = Mdx + Ndy$  ..... ②

Again since,  $f = f(x, y)$ ,  $\Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$  ..... ③

② & ③  $\Rightarrow Mdx + Ndy = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$

$$\therefore M = \frac{\partial f}{\partial x}$$

$$N = \frac{\partial f}{\partial y}$$

$$\begin{aligned} \text{Now, } \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} \\ &= \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial y} \right) \\ &= \frac{\partial N}{\partial x} \end{aligned}$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

That the conditions that is necessary has been proved.

• Sufficient condition :-

We prove that if  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ , then  $Mdx + Ndy = 0$  is an exact eq<sup>n</sup>.

Let,  $F = \int Mdx$  then  $\frac{\partial F}{\partial x} = M$  ..... ④

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Now,  $\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x} = \frac{\partial}{\partial y} \left( \frac{\partial F}{\partial x} \right) = \frac{\partial M}{\partial y}$  using (4)

Again,  $\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial M}{\partial y}$

$$\Rightarrow \frac{\partial^2 F}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

$$\Rightarrow \frac{\partial N}{\partial x} - \frac{\partial^2 F}{\partial x \partial y} = 0$$

$$\Rightarrow \frac{\partial}{\partial x} \left( N - \frac{\partial F}{\partial y} \right) = 0$$

$$\Rightarrow N - \frac{\partial F}{\partial y} = \phi(y), \text{ where } \phi(y) \text{ is a function of } y \text{ free } x.$$

$$\Rightarrow N = \frac{\partial F}{\partial y} + \phi(y)$$

Now,  $Mdx + Ndy = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy + \phi(y) dy$

$$= dF + \phi(y) dy$$

$$= dF + d \left( \int \phi(y) dy \right)$$

$$= d \left[ F + \int \phi(y) dy \right]$$

$$= df$$

Thus the condition is sufficient has been proved.

□ If the equation is exact then  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$   $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$   
 $\therefore$  the solution of the equation is  $\int M dx + \int N(x\text{-free}) dy = C$

①  $(3x + 2y - 5) dx + (2x + 3y - 5) dy = 0 \dots \dots \dots$  ①

Sol<sup>n</sup>:-

Here,  $M = 3x + 2y$  ;  $N = 2x + 3y$

$$\Rightarrow \frac{\partial M}{\partial y} = 2 \quad ; \quad \Rightarrow \frac{\partial N}{\partial x} = 2$$

Since,  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ , so the equation is exact.

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Hence, the solution is,

$$\int (3x + 2y - 5) dx + \int (3y - 5) dy = c_1$$

$$\Rightarrow \frac{3x^2}{2} + 2xy - 5x + \frac{3}{2}y^2 - 5y = c_1$$

$$\Rightarrow 3x^2 + 4xy + 3y^2 - 10x - 10y = 2c_1$$

$$\Rightarrow 3x^2 + 4xy + 3y^2 - 10x - 10y = c$$

$$\textcircled{2} (x^3 + 3xy^2) dx + (y^3 + 3x^2y) dy = 0$$

Here,  $M = x^3 + 3xy^2$  ,  $N = y^3 + 3x^2y$

$$\frac{\partial M}{\partial y} = 6xy$$

$$\frac{\partial N}{\partial x} = 6xy$$

Since,  $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$  , so the equation is exact.

Hence the solution is,  $\int (x^3 + 3xy^2) dx + \int y^3 dy = c_1$

$$\Rightarrow \frac{x^4}{4} + \frac{3}{2}x^2y^2 + \frac{y^4}{4} = c_1$$

$$\Rightarrow x^4 + 6x^2y^2 + y^4 = 4c_1 = c_2$$

$$\textcircled{3} (1 + e^{x/y}) dx + e^{x/y} (1 - x/y) dy = 0$$

Here,  $M = 1 + e^{x/y}$  ;  $N = e^{x/y} (1 - x/y)$

$$\Rightarrow \frac{\partial M}{\partial y} = -\frac{x}{y^2} e^{x/y}$$

$$\begin{aligned} \frac{\partial N}{\partial x} &= e^{x/y} \frac{1}{y} (1 - \frac{x}{y}) - e^{x/y} (0 - \frac{1}{y}) \\ &= \frac{e^{x/y}}{y} - \frac{xe^{x/y}}{y^2} + \frac{e^{x/y}}{y} \\ &= -\frac{xe^{x/y}}{y^2} \end{aligned}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\Rightarrow \int (1 + e^{x/y}) dx + \int 0 dy = c$$

$$\Rightarrow x + ye^{x/y} = c$$

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\* if  $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ , M and N both are homogeneous and  $Mx + Ny \neq 0$   
 $\therefore I.F = \frac{1}{Mx + Ny}$

\* if  $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ , M and N are not homogeneous and  $Mx - Ny \neq 0$   
 $\therefore I.F = \frac{1}{Mx - Ny}$

$$\textcircled{1} (x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0 \dots\dots\dots \textcircled{1}$$

Sol<sup>n</sup>:- Here,  $M = x^2y - 2xy^2$        $N = -x^3 + 3x^2y$

$$\frac{\partial M}{\partial y} = x^2 - 4xy \qquad \frac{\partial N}{\partial x} = 3x^2 + 6xy$$

Since,  $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$  and M & N are homogeneous

$$I.F = \frac{1}{Mx + Ny} = \frac{1}{x^2y^2}$$

Multiplying  $\textcircled{1}$  by I.F we get,

$$\left(\frac{1}{y} - 2\frac{1}{x}\right)dx - \left(\frac{x}{y^2} - 3\frac{1}{y}\right)dy = 0$$

Hence the solution is,  $\int \left(\frac{1}{y} - \frac{2}{x}\right)dx + \int \frac{3}{y}dy = c$   
 $\Rightarrow \frac{x}{y} - 2\ln x + 3\ln y = c$

$$\textcircled{2} (1+xy)ydx + (1-xy)x dy = 0 \dots\dots\dots \textcircled{1}$$

Here,  $M = (1+xy)y$        $N = (1-xy)x$

$$\frac{\partial M}{\partial y} = 1 + 2xy \qquad \frac{\partial N}{\partial x} = 1 - 2xy$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$Mx - Ny = xy + x^2y^2 - xy + x^2y^2 = 2x^2y^2$$

$$\therefore I.F = \frac{1}{2x^2y^2}$$

① can be written as,

$$\int \frac{y + xy^2}{2x^2y^2} dx + \int \frac{x - x^2y}{2x^2y^2} dy = 0$$

$$\left(\frac{1}{2x^2y} + \frac{1}{2x}\right) dx + \left(\frac{1}{2xy^2} - \frac{1}{2y}\right) dy = 0$$

Hence the solution is,

$$\int \left(\frac{1}{2x^2y} + \frac{1}{2x}\right) dx + \int \frac{1}{2y} dy = c$$

$$\Rightarrow \int \left(\frac{1}{x^2y} + \frac{1}{x}\right) dx - \int \frac{1}{y} dy = c$$

$$\Rightarrow -\frac{1}{xy} + \ln x - \ln y = c$$

কি If  $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ ,  $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}$  কে যদি  $N$  দ্বারা ভাগ করা যায় এবং যেখানে  
যদি সুস্থিতি  $x$  এর function পাওয়া যায় তবে,

$$I.F = e^{\int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) dx}$$

যদি  $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$  কে  $M$  দ্বারা ভাগ করলে সুস্থিতি  $y$  এর function পাওয়া  
যায় তবে,

$$I.F = e^{\int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dy}$$

$$\textcircled{1} (12y + 4y^3 + 6x^2) dx + 3(x + xy^2) dy = 0 \dots \dots \dots \textcircled{1}$$

Sol<sup>n</sup>:- Here,  $M = 12y + 4y^3 + 6x^2$

$$N = 3(x + xy^2)$$

$$\frac{\partial M}{\partial y} = 12 + 12y^2$$

$$\frac{\partial N}{\partial x} = 3(1 + y^2)$$

Since,  $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ , so the equation is not exact.

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$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 9 + 9y^2$$

$$\frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{9(1+y^2)}{3x(1+y^2)}$$

$$\Rightarrow \frac{1}{N} \left( \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{3}{x}$$

$$I.F = e^{\int \frac{3}{x} dx} = x^3$$

$$\textcircled{1} \text{ become } (12yx^3 + 4x^3y^3 + 6x^5)dx + 3(x^4 + 3x^4y^2)dy = 0$$

$$\text{Hence the solution is } 3x^4y + x^4y^3 + x^6 = c.$$

$$\textcircled{2} y \log y dx + (x - \ln y) dy = 0 \dots \dots \dots \textcircled{1}$$

$$\text{Here, } M = y \log y \quad N = x - \ln y$$

$$\frac{\partial M}{\partial y} = 1 + \ln y \quad \frac{\partial N}{\partial x} = 1$$

$$\text{Since, } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -\ln y$$

$$\frac{1}{M} \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = -\frac{1}{y} \text{ which is a function of } y.$$

$$I.F = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

$$\textcircled{1} \text{ Can be written as } \log y dx + \left( \frac{x}{y} - \frac{\ln y}{y} \right) dy$$

$$\text{Hence, the solution is, } \int \log y dx + \int -\frac{\ln y}{y} dy = c$$

$$\Rightarrow x \ln y - \frac{(\ln y)^2}{2} = c.$$